

PegasusX — Order Flow & Edge Cases (CODE)

SOURCE OF TRUTH:

- Statuses: `apps/backend-go/order/service.go` (`Status*` constants)
- Transitions: `apps/backend-go/order/state_machine.go` (`ValidateStatusTransition`)
- Triggers: `apps/backend-go/order/routes/routes.go` , `driver/routes/routes.go` , `retailer/routes/routes.go`
- Tests: `order/state_machine_test.go` , `simulator/order_lifecycle_test.go` , `order/fiscal_test.go`

Not derived from other markdown.

1. Status constants (`order/service.go`)

```
PENDING
LOADED
IN_TRANSIT
ARRIVED
SHOP_CLOSED_PENDING
AWAITING_PAYMENT
PENDING_CASH_COLLECTION
DELIVERED_ON_CREDIT
FISCALIZING          // ADR-009: payment captured, OFD in flight
FISCAL_FAILED
COMPLETED
CANCELLED
CANCEL_REQUESTED
RECONCILIATION_REQUIRED
DELAYED
BACKORDERED
SCHEDULED
AUTO_ACCEPTED
```

ConfirmationStatus: `CONFIRMED` , `DRAFT` , `PENDING` , `REJECTED` , `AUTO_CONFIRMED` , `PENDING_WAREHOUSE`

OrderSource: `MANUAL` , `MANUAL_PREORDER` , `AI_PREORDER` , `BACKORDER` , `AUTO_ORDER`

Sentinel errors (subset): `invalid_status_transition` , `geofence_violation` , `zone_miss` , `inventory_exhausted` , `credit_limit_breached` , `proximity_required` , `force_complete_forbidden` , `cash_amount_required` , ...

2. Exact transition table (`ValidateStatusTransition`)

Same from == to → allow (idempotent). Otherwise:

From	Allowed to (code)
PENDING	LOADED, CANCELLED, DELAYED
LOADED	IN_TRANSIT, CANCELLED, CANCEL_REQUESTED, DELAYED, PENDING
DELAYED	PENDING
IN_TRANSIT	ARRIVED, CANCELLED, CANCEL_REQUESTED, PENDING
ARRIVED	AWAITING_PAYMENT, PENDING_CASH_COLLECTION, DELIVERED_ON_CREDIT, CANCEL_REQUESTED, SHOP_CLOSED_PENDING
SHOP_CLOSED_PENDING	AWAITING_PAYMENT, PENDING_CASH_COLLECTION, DELIVERED_ON_CREDIT, CANCELLED, ARRIVED, CANCEL_REQUESTED
DELIVERED_ON_CREDIT	FISCALIZING
AWAITING_PAYMENT	FISCALIZING, PENDING_CASH_COLLECTION, DELIVERED_ON_CREDIT
PENDING_CASH_COLLECTION	FISCALIZING
FISCALIZING	COMPLETED, FISCAL_FAILED
FISCAL_FAILED	FISCALIZING, COMPLETED
CANCEL_REQUESTED	CANCELLED, LOADED, IN_TRANSIT, ARRIVED
COMPLETED	(none)
CANCELLED	RECONCILIATION_REQUIRED
RECONCILIATION_REQUIRED	COMPLETED, CANCELLED
BACKORDERED	PENDING, SCHEDULED, CANCELLED
SCHEDULED	AUTO_ACCEPTED, PENDING, CANCELLED, CANCEL_REQUESTED
AUTO_ACCEPTED	PENDING, CANCELLED, CANCEL_REQUESTED
default	deny

Comments in `state_machine.go` :

- ADR-009: no soft ARRIVED → COMPLETED
- Credit leave only reaches COMPLETED via FISCALIZING after money
- CANCEL_REQUESTED must have exits or it bricks the order

```
stateDiagram-v2
    [*] --> PENDING
    PENDING --> LOADED
    PENDING --> CANCELLED
    PENDING --> DELAYED
    DELAYED --> PENDING
    LOADED --> IN_TRANSIT
    LOADED --> CANCELLED
    LOADED --> CANCEL_REQUESTED
    LOADED --> DELAYED
    LOADED --> PENDING
    IN_TRANSIT --> ARRIVED
```

```

IN_TRANSIT --> CANCELLED
IN_TRANSIT --> CANCEL_REQUESTED
IN_TRANSIT --> PENDING
ARRIVED --> AWAITING_PAYMENT
ARRIVED --> SHOP_CLOSED_PENDING
ARRIVED --> PENDING_CASH_COLLECTION
ARRIVED --> DELIVERED_ON_CREDIT
ARRIVED --> CANCEL_REQUESTED
SHOP_CLOSED_PENDING --> AWAITING_PAYMENT
SHOP_CLOSED_PENDING --> PENDING_CASH_COLLECTION
SHOP_CLOSED_PENDING --> DELIVERED_ON_CREDIT
SHOP_CLOSED_PENDING --> CANCELLED
SHOP_CLOSED_PENDING --> ARRIVED
SHOP_CLOSED_PENDING --> CANCEL_REQUESTED
AWAITING_PAYMENT --> FISCALIZING
AWAITING_PAYMENT --> PENDING_CASH_COLLECTION
AWAITING_PAYMENT --> DELIVERED_ON_CREDIT
PENDING_CASH_COLLECTION --> FISCALIZING
DELIVERED_ON_CREDIT --> FISCALIZING
FISCALIZING --> COMPLETED
FISCALIZING --> FISCAL_FAILED
FISCAL_FAILED --> FISCALIZING
FISCAL_FAILED --> COMPLETED
CANCEL_REQUESTED --> CANCELLED
CANCEL_REQUESTED --> LOADED
CANCEL_REQUESTED --> IN_TRANSIT
CANCEL_REQUESTED --> ARRIVED
CANCELLED --> RECONCILIATION_REQUIRED
RECONCILIATION_REQUIRED --> COMPLETED
RECONCILIATION_REQUIRED --> CANCELLED
BACKORDERED --> PENDING
BACKORDERED --> SCHEDULED
BACKORDERED --> CANCELLED
SCHEDULED --> AUTO_ACCEPTED
SCHEDULED --> PENDING
SCHEDULED --> CANCELLED
AUTO_ACCEPTED --> PENDING
AUTO_ACCEPTED --> CANCELLED
COMPLETED --> [*]

```

3. Who triggers what (mounted routes)

3.1 Create / early lifecycle

Action	Route	Roles (from RequireRole)
Create	POST /v1/order/create	RETAILER
Checkout	POST /v1/checkout/{unified,preview,b2b}	paymentroutes

Action	Route	Roles (from RequireRole)
Cash/card checkout	POST /v1/order/{cash,card}-checkout	retailerroutes
Patch status	PATCH /v1/order/{orderId}/status	ADMIN, RETAILER
Assign	POST /v1/orders/{orderId}/assign	ADMIN, WAREHOUSE_ADMIN, FACTORY_ADMIN
Supplier vet	POST /v1/supplier/orders/vet	ADMIN
WH delay/reject/overflow/propose	/v1/warehouse/ops/orders/{id}/*	WAREHOUSE_ADMIN, ADMIN

3.2 Doorstep (orderroutes + driverroutes)

Action	Route	Role
Arrive	POST /v1/delivery/arrive	DRIVER
Proximity unlock	POST /v1/delivery/proximity-unlock	DRIVER
Shop closed	POST /v1/delivery/shop-closed (+ driver alias .../orders/{id}/shop-closed)	DRIVER
Partial offload	POST /v1/delivery/partial-offload (+ driver alias)	DRIVER
Scan QR	POST /v1/delivery/scan-qr	DRIVER
Deliver	POST /v1/order/deliver	DRIVER
Confirm offload	POST /v1/order/confirm-offload	DRIVER
Confirm cash	POST /v1/delivery/confirm-cash	RETAILER
Collect cash	POST /v1/order/collect-cash	DRIVER
Complete	POST /v1/order/complete	DRIVER
Fiscal retry	POST /v1/order/{id}/fiscal/retry	DRIVER, ADMIN, WAREHOUSE_ADMIN
Force complete	POST /v1/order/{id}/force-complete	ADMIN, WAREHOUSE_ADMIN
Credit leave	POST /v1/driver/orders/{id}/credit-leave , POST /v1/delivery/credit-delivery	DRIVER
Bypass / payment bypass confirm	POST /v1/delivery/bypass-offload , confirm- payment-bypass	DRIVER
Damage / condition	POST /v1/delivery/report-damage , report- condition	DRIVER (+ others on condition)
Missing / exception / split / negotiate	driverroutes delivery*	DRIVER
Sync batch	POST /v1/sync/batch	DRIVER
Retailer shop- closed	POST /v1/retailer/shop-closed-response , .../orders/{id}/shop-closed/respond	RETAILER

Action	Route	Role
Supplier shop-closed resolve	POST /v1/supplier/shop-closed/resolve	ADMIN

3.3 Post-delivery

Action	Route	Role
File claim	POST /v1/orders/{id}/claims	RETAILER, ADMIN
Eligibility / list	GET claim-eligibility, claims	RETAILER, ADMIN, WAREHOUSE_ADMIN (list)
Approve / reject claim	POST /v1/claims/{id}/{approve,reject}	ADMIN, WAREHOUSE_ADMIN
Reconciliation	/v1/supplier/reconciliation*	ADMIN

4. Happy paths (composed from transitions + routes)

4.1 Cash COD

1. Checkout → PENDING (+ ReserveLineItemsInTxn)
2. Dispatch/seal/depart → LOADED → IN_TRANSIT
3. POST /v1/delivery/arrive → ARRIVED
4. Scan/deliver/confirm-offload → AWAITING_PAYMENT (or cash choice → PENDING_CASH_COLLECTION)
5. POST /v1/order/collect-cash → FISCALIZING (+ cash variance if expected≠received)
6. Fiscal worker SUCCESS → COMPLETED

4.2 Card

Same through AWAITING_PAYMENT → POST /v1/order/complete → FISCALIZING → COMPLETED .

4.3 Credit leave

ARRIVED / AWAITING_PAYMENT / SHOP_CLOSED_PENDING → DELIVERED_ON_CREDIT (gate CanLeaveOnCredit) → later money capture → FISCALIZING → COMPLETED .

Illegal: DELIVERED_ON_CREDIT → COMPLETED directly (state_machine.go).

4.4 Preorder

Create → SCHEDULED (MANUAL_PREORDER / AI_PREORDER). Sweeper may → AUTO_ACCEPTED .
Promote → PENDING . Then §4.1.

4.5 Backorder

BACKORDERED → PENDING or SCHEDULED or CANCELLED (no reserve path for backorder source in reservation guards).

5. Edge cases (asserted or implemented in code)

5.1 State machine

Edge	Behavior	Where
Idempotent same status	OK	ValidateStatusTransition
COMPLETED → anything	Denied	state_machine + tests
Soft complete from ARRIVED/AWAITING/PENDING_CASH	Denied	ADR-009 comment + state_machine_test.go
Credit → COMPLETED	Denied	state_machine + simulator
CANCEL_REQUESTED exits	CANCELLED or resume LOADED/IN_TRANSIT/ARRIVED	state_machine
LOADED/IN_TRANSIT → PENDING	Allowed (rollback)	state_machine + simulator

5.2 Inventory

Edge	Behavior	Where
qoh - qr < qty	inventory_exhausted	ReserveLineItemsInTxn
Duplicate SKUs	Aggregated before write	same
Missing SKU row	exhausted	same
Stale PENDING	Cancel path release	inventory_stale_release.go
SCHEDULED without marker	Backfill reserve	BackfillScheduledReservations

5.3 Geofence / proximity

Edge	Behavior	Where
Approach radius	500 m constant	proximity/geofence.go
Settlement	unlock or live ~100 m / H3 (service comment)	order/service.go CollectCash area
Missing proximity	proximity_required	service errors

5.4 Cash / fiscal

Edge	Behavior	Where
Variance	shortfall/overage events	emitCashVariance
OFD timeout	8s	FiscalOFDTimeout
Retry	FISCAL_FAILED → FISCALIZING	fiscal/retry route

Edge	Behavior	Where
Force-complete roles	ADMIN, WAREHOUSE_ADMIN only	orderroutes
Force from credit leave	Not via DELIVERED_ON_CREDIT→COMPLETED	state_machine comment
Idempotent collect-cash	key + body hash	idempotency_guard.go

5.5 Shop closed

Edge	Behavior	Where
Report	ARRIVED → SHOP_CLOSED_PENDING	shop-closed handlers
Default grace	5 minutes if unset	NewService
Timeout credit	CanLeaveOnCredit + MaxAutoCredit 50_000_000	worker_shop_closed.go
Retailer respond	routes in retaileroutes	retailer_shop_closed.go
Bypass	TransitionOpts PhotoURL / SupervisorToken	state_machine opts + bypass route

5.6 Partial offload

Edge	Behavior	Where
Qty mismatch	partial_qty_mismatch	ApplyPartialOffloadLines
Invalid status	partial_invalid_status	apply path status guard
Totals	delivered_minor / remaining_minor	partial_offload.go

5.7 Cancel

Edge	Behavior	Where
request-cancel / cancel	retailerroutes	cancel_service.go
Mid-leg	CANCEL_REQUESTED then approve/resume	state_machine
CANCELLED → recon	RECONCILIATION_REQUIRED	state_machine

5.8 Claims

Edge	Behavior	Where
Only COMPLETED	eligibility	claims/eligibility.go
Default window 48h	const	claims/service.go
Price from order lines	aggregate + half-up avg	claims/pricing.go
Prior qty statuses	OPEN/UNDER_REVIEW/APPROVED/RESOLVED	ClaimedQtyBySKU

5.9 Tests to re-run when changing graph

- order/state_machine_test.go
- simulator/order_lifecycle_test.go

- `order/fiscal_test.go`
 - `order/shop_closed_timeout_test.go`
-

6. Side fields on `Order` affecting flow

From repository/select columns (non-exhaustive): `FiscalStatus` , fiscal receipt/attempt ids, `ShopClosedAt/Reason/GraceEndsAt/Resolution` , `PartialDelivery` , `ProximityUnlockedAt/Method` , claim window fields, preorder lock/reminder fields, delivery proposal fields, `Version` (optimistic concurrency).

7. Rebuild rule

1. Diff against `ValidateStatusTransition` switch — table must match line-for-line.
2. Add triggers only if present in `*routes/routes.go` with `RequireRole` .
3. Do not copy edge lists from other docs.